

Camera-Derived Road Context for Physically Interpretable Vehicle Prediction

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Abstract

Physically meaningful predictive states make learned vehicle models easier to inspect, but road-aligned coordinates introduce geometric and numerical assumptions. We extend the Physically Interpretable World Model (PIWM) framework with a camera-derived finite road preview and structured road-relative dynamics. The task is offline prediction conditioned on a known initial state and recorded future actions. A road encoder predicts curvature and centerline shape from a 15-frame camera history; the controlled evaluation reconstructs motion without querying the surveyed centerline during rollout. Initial road-relative state remains privileged. We audit the formulation on two existing physical DonkeyCar recordings, holding out each recording in turn and separating checkpoint selection from evaluation. Matched controls include calibrated kinematics and Cartesian residual models with and without the same visual road inputs. The initial Frenet implementation gives 100-step endpoint errors of 1.086 and 1.463 m, compared with 0.936 and 1.158 m for calibrated kinematics. Exploratory follow-ups diagnose unstable coordinate updates, inconsistent curvature and shape parameterizations, and one nearly constant visual encoder. In a subsequent purged chronological holdout of both sources, the fixed guarded model achieves 0.343 m pooled endpoint error, versus 0.898 m for calibrated kinematics and 1.095 m for the same-input Cartesian control. These improvements are specific to the chronological setting; the source-transfer tests do not establish a consistent advantage from curvature-dependent dynamics. The contribution is an explicit road representation together with a reproducible assessment of its information requirements, geometric consistency, and empirical limits. Repeated development on the same recordings constrains generalization claims; neither unseen-track transfer nor camera-only deployment is demonstrated.

Keywords: physically interpretable prediction, road geometry, vehicle dynamics, visual representation learning, retrospective evaluation

1. Introduction

World models compress observations into states that can be propagated under candidate actions [1, 2]. Physical interpretability provides a useful complement to predictive accuracy: a predicted quantity should have an identifiable meaning and unit, and its transition should expose the assumptions connecting it to the modeled system. Our conference PIWM framework [3] investigated physical alignment under distribution-based weak supervision. This article studies a road-context extension on physical DonkeyCar records.

Road geometry matters for interpreting lateral displacement and heading relative to a lane, and for the actions chosen by a driving controller. It does not, by itself, change planar motion when the complete vehicle state, actuation sequence, and physical conditions are fixed. We ask whether explicit visual road context provides a useful inductive bias for finite-horizon action-conditioned prediction, rather than claiming that identical physical states and actions must produce different motion on differently painted roads.

Our state contains speed, yaw rate, progress, lateral offset, and road-relative heading, accompanied by a finite road preview. The preview is perceived once and queried at predicted progress. This avoids recurrent prediction of the profile, but does not prevent errors in progress, perception, or actuation from propagating. The Frenet chart can become singular, and independently predicted curvature and centerline shape need not describe the same curve.

Two original recordings on one laboratory track provide 71 usable segments, not 71 independent acquisition sessions. The new evaluations separate fitting and validation within one source and evaluate on the other. Both sources had already informed development, so this analysis is retrospective. Earlier five-fold results are preserved separately because their information and model-selection conditions differ from those of the new comparisons. A subsequent purged temporal split includes both sources in training and evaluates their later segments, while retaining all source-transfer results.

The extensions beyond the conference study are: (i) a camera-derived curvature and centerline preview with explicit road-relative state and map-free rollout readout; (ii) analysis of coordinate validity, geometric consistency, and

visual optimization; and (iii) controlled source-separated and chronological experiments distinguishing numerical stability from accuracy. The Frenet equations themselves are standard coordinate transformations, not a new physical law. The evidence supports an inspectable modeling approach and identifies limitations; it does not establish universal superiority over calibrated dynamics.

Sections 2 and 3 provide background. Sections 4–5 specify the implemented task and models; Sections 6–7 describe the protocol and findings, followed by limitations and conclusions.

2. Related Work

We organize the discussion into five threads relevant to physically interpretable world modeling under partial observability: trajectory prediction and safe forecasting (Section 2.1), representation learning for dynamical systems (Section 2.2), world models (Section 2.3), physics-informed and interpretable learning (Section 2.4), and partial observability, privileged learning, and road structure (Section 2.5). We close each thread by positioning PIWM against it.

2.1. Trajectory Prediction and Safe Forecasting

Predicting future motion is central to safe planning and control [4]. Classical predictors propagate a low-dimensional state with hand-specified models: Kalman-filter formulations over a bicycle model of vehicle motion [5], and Hamilton–Jacobi reachability for sets of reachable states with formal guarantees [6, 7, 8]. These approaches expose low-dimensional modeling assumptions. Their use with camera observations requires an appropriate state-estimation interface, and any formal guarantee depends on the assumptions of the particular method.

Learning-based predictors instead forecast directly from data. Social-pooling and generative sequence models capture multi-agent interaction [9, 10, 11], while graph- and attention-based models such as Trajectron++ achieve strong forecasting by conditioning on scene graphs and high-definition maps [12]. Hybrid schemes embed classical filters inside learned models, as in learned Kalman filtering [13]. Scene representations and map inputs differ across these methods; availability of suitable annotations is one practical consideration [14, 15]. Physical interpretability must be assessed for each representation rather than inferred from whether a model is learned.

A parallel line equips predictors with reliability guarantees without altering the predictor itself: conformal prediction and calibrated forecasting bound error or coverage [16, 17, 18, 19]; statistical model checking and predictive monitoring estimate the probability of property violations [20, 21, 22, 23]; and runtime and misbehavior monitors flag unsafe behavior online [24, 25, 26, 27]. These techniques are complementary to ours: they wrap a predictor with reliability analyses under their own assumptions; this is distinct from assigning physical semantics to a predictor’s internal coordinates. Our driving study examines named road variables and structured transitions. Its controlled evaluation retains ground-truth initial state and direct map-derived training labels; it does not demonstrate fully unaided visual driving.

2.2. Representation Learning for Dynamical Systems

Compact latent representations underpin learning from high-dimensional observations, from classical autoencoders [28] to variational autoencoders (VAEs) [29], but standard objectives impose no physically meaningful structure. A large literature seeks more structured latents by encouraging disentanglement [30, 31, 32], causal structure [33, 34], or object-centric and slot decompositions [35]. Discrete VQ-VAEs regularize the latent through a learned codebook [36, 37]; the conference PIWM formulation investigates physically supervised quantized representations. The current road encoder is continuous. These methods improve organization or modularity, yet none guarantees correspondence to measurable physical quantities.

Closer to physical grounding, several models steer latents toward physics. GOKU-net constrains the latent dynamics to a known ODE and infers that ODE’s parameters [38], but the functional form must be fixed a priori and the latents are tied to it rather than to quantities that can be measured from the observation; SPARTAN produces semantically structured outputs but incorporates neither known dynamics nor action-conditioned prediction [39]; Deep Variational Bayes Filters add latent state-space dynamics to VAEs and report latents that correlate with physical quantities [40], but the correspondence emerges from the transition structure rather than being enforced, so which dimension carries which quantity is not fixed in advance; and continuous-time models such as Neural ODEs learn smooth but uninterpretable latent dynamics [41]. Representation learning specialized for pixel-based control has been studied extensively for its effect on downstream performance [42], while again leaving the latent opaque. The conference PIWM study uses

distribution-based supervision. The driving extension instead uses direct road labels to evaluate a manually specified physical representation.

2.3. *World Models*

World models learn latent dynamics for imagination-based planning and control. The Dreamer and PlaNet line learns recurrent latent state-space models from pixels [43, 44, 2]; transformer- and masked-prediction world models improve sample efficiency [45, 46], and the same family has been trained directly on physical robots [47]; and a range of variants explore alternative latent factorizations [1, 48, 49]. These models excel empirically but learn latent spaces with no clear physical meaning, which limits inspection and safety monitoring.

For driving, recent world models predict rich structured futures such as 3D occupancy grids or rendered scenes [50, 51, 52, 53], improving forecasting realism but still without aligning the internal state to physical variables. Neuro-symbolic world models improve generalization, but their symbolic vocabulary is either supplied in advance or induced separately from the physical quantities a controller acts on [54, 55]. Vid2Param estimates physical parameters from video [56]. The new comparisons here use explicitly specified controls rather than claiming full reproductions of these published architectures.

2.4. *Physics-Informed and Interpretable Learning*

A complementary thread injects physics directly into learning. Sparse identification of nonlinear dynamics and differentiable-physics or end-to-end system identification recover governing equations or parameters [57, 58, 59], while physics-informed neural networks embed differential constraints in the loss [60, 61]. Within world modeling, physical priors have been added as bounds on states and actions [62, 63], as known physics entering the model’s drift term as an inductive bias [64], kinematics-inspired layers [65], and Taylor-monomial-structured latent dynamics (Phy-Taylor) [66].

Most of these methods assume access to the underlying low-dimensional state and are not designed for raw visual inputs, or they enforce physical plausibility without yielding a state that names specific quantities. We combine image-derived road geometry with road-relative pose kinematics and learned actuation increments. This inductive bias requires comparison with controls having comparable inputs and training conditions.

2.5. *Partial Observability, Privileged Learning, and Road Structure*

Partial observability is a long-standing challenge in robotics and reinforcement learning, classically formalized through partially observable Markov decision processes and belief-state planning [67, 68]. Model-free agents commonly cope with it by adding memory through recurrent or temporal architectures: recurrent policy gradients [69], gated recurrent networks [70], convolutional LSTMs for spatiotemporal inputs [71], and temporal convolutional networks [72]. These approaches recover an opaque belief or memory state; they do not produce a physically interpretable estimate of the missing environmental variables, which is exactly what our road-context state provides.

When information needed for learning is available only at training time, privileged-information and teacher–student learning supervise a model that must operate without it at deployment [73, 74]. Our preview encoder follows this training pattern, but rollout initialization uses annotated pose and track geometry. Privileged information is therefore reduced during prediction, not eliminated from the complete evaluation.

In driving specifically, a substantial literature encodes lane and road structure in learned representations [12, 14], but typically requires explicit maps or HD annotations; and physics-based predictors apply kinematic or dynamic bicycle models to low-dimensional state vectors [75] rather than to learned image representations. Learning-based control has also been demonstrated on RC-scale physical cars, including reinforcement learning on the DonkeyCar platform [76], but without an interpretable predictive world model. Our focus is the empirical role of explicit road context under action-conditioned prediction, including numerical limitations. We do not claim priority for combining lane geometry and vehicle physics in general.

Frenet-based trajectory representations predate the present study. Ye et al. [77] use Frenet domain normalization to address scene shift, while Hallgarten et al. [78] represent prediction inputs and outputs relative to lane-centerline sequences to reduce off-road forecasts. Their behavior-prediction setting differs from our supplied-action vehicle rollout, so their reported scores are not directly comparable. More recently, GeoWAM [79] forecasts visual scene geometry and conditions ego-trajectory prediction on it. This is a broader geometric scene representation than our compact road preview. These connections motivate a specific contribution claim rather than claiming that road-aligned coordinates or explicit geometric world states are themselves new.

3. Preliminaries

3.1. Autonomous CPS and World Models

Definition 1 (Autonomous CPS). An autonomous CPS, represented by the tuple $s = (X, I, Y, A, \phi_\theta, g, h)$, models the evolution of an observation-based control system, where X is the physical state space, $I \subset X$ the set of initial states, Y the observation space, A the action space, $\phi_\theta : X \times A \times \Theta \rightarrow X$ the dynamics with unknown parameters $\theta \in \Theta$, $g : X \rightarrow Y$ the observation function, and $h : Y \rightarrow A$ the controller.

The true state evolves as $x_{t+1} = \phi(x_t, a_t, \theta^*)$, where θ^* are the unknown true parameters. The controller selects actions from observations: $a_t = h(y_t) = h(g(x_t))$. A world model serves as a learned surrogate for forecasting future observations.

Definition 2 (World Model). A world model $\mathcal{W} = (\mathcal{E}, f, \mathcal{D})$ predicts the future evolution of observations in an autonomous CPS s , where $\mathcal{E} : Y \rightarrow Z$ encodes observations into latent representations, $f : Z \times A \rightarrow Z$ predicts future latents, and $\mathcal{D} : Z \rightarrow Y$ reconstructs predicted observations.

Conventional world models are evaluated by predictive accuracy in image space (e.g., MSE or SSIM). However, low image error does not imply that the underlying predicted state is physically meaningful: similar-looking frames can correspond to very different physical configurations. For safety monitoring and planning, what matters is not only visual fidelity but whether the latent state can be interpreted as a physical quantity and evolved according to physically plausible dynamics.

3.2. Physical Interpretability

Following the conference paper [3], we define *physical interpretability* via two complementary properties:

1. the latent representation z^* corresponds to meaningful physical variables (state semantics), and
2. its temporal evolution follows physically consistent dynamics (temporal correctness).

Property 1 requires alignment between the latent and the physical state, which PIWM enforces through weak supervision. Property 2 requires that multi-step rollouts of the latent state obey the known equations of motion, which PIWM encourages through its structured dynamics prior; learned residuals and numerical integration still require empirical checks.

Definition 3 (Interpretable Representation Learning Problem). *Given a dataset $\mathcal{S} = \{(y_{1:M}, a_{1:M}, \Xi_{1:M})\}_{1:N}$, where $\Xi_t = \{\xi_t^{(l)}\}_{l=1}^L$ are samples drawn from an unknown supervisory distribution $p(x_t)$, and a controller h , learn a latent encoder $\mathcal{E}$ such that the mean squared error between the physical state x_t and the encoded latent $z_t^* = \mathcal{E}(y_t)$ is minimized:*

$$\min_{\mathcal{E}} \mathbb{E}[\|x_t - \mathcal{E}(y_t)\|_2^2]. \quad (1)$$

The supervisory samples Ξ_t are drawn from an unknown distribution $p(x_t)$ with no assumed parametric form; we only require that the distribution is *mean-proximal*, meaning its mean lies within a relative interval of width δ around the true value (see Section 6 for the precise noise model). This is a strictly weaker assumption than standard supervised learning with precise labels, and represents the uncertainty naturally arising from real-world sensing.

These definitions describe the conference framework. In the driving study, the visual output is a road preview, not a reconstructed image, and the initial vehicle state is supplied. Exact geometric supervision here differs from the conference weak-supervision setting.

4. Prediction Task and Information Conditions

At the beginning of each offline prediction window, the inputs are a 15-frame camera history, initial speed v_0 , a causal yaw-rate estimate ω_0 , initial road-relative displacement d_0 and heading error ψ_0 , and recorded future steering/throttle actions. The output is planar motion in the initial ego frame. Future images, future measured states, and controller feedback are not supplied. Recorded actions are conditioning inputs, not predicted controls.

4.1. Factorized state

The state is $z = (s, d, \psi, v, \omega)$, accompanied by curvature samples and centerline points at ten offsets from 0 to 4.5 m. Initial progress is zero. Road context defines the coordinate system; we do not assert that this finite preview is a sufficient Markov state for arbitrary driving environments.

The surveyed track and logged pose provide road labels and initial d_0 and ψ_0 . The controlled forecast readout uses predicted road points instead of the surveyed centerline. Evaluation uses original logged planar positions, not positions reconstructed from projected road labels. Thus label construction, privileged initialization, forecast readout, and evaluation truth are distinct. This is not a complete camera-only estimator or closed-loop driving system.

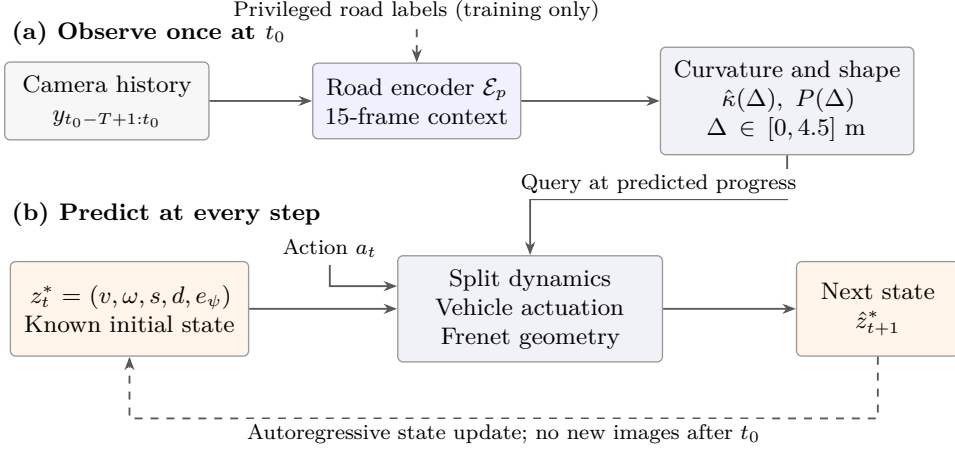


Figure 1: Implemented prediction pathways. The camera supplies curvature and shape once per rollout; annotated state initializes the dynamics. Recorded actions drive subsequent updates. The original and guarded variants query the curvature head; the consistent variant derives curvature and parameter speed from the shape curve. Position is decoded using that predicted curve, without surveyed-map queries during rollout. Initial road-relative state remains privileged.

5. Implemented Driving Models

5.1. Road-preview encoder

A convolutional encoder receives the red channel of 15 consecutive 64×64 frames, stacked as channels and scaled to $[0, 1]$. Four convolutions have 32, 64, 128, and 256 channels, kernel size 4, stride 2, and padding 1. The flattened output feeds a 256-unit fully connected layer. This feature vector feeds linear heads for ten curvature samples and twenty centerline coordinates. The longitudinal shape output is a residual relative to nominal offsets; the lateral output is direct. Joint supervised squared error is normalized by training-set output standard deviations, floored at 0.05 in the corresponding units. The controlled experiment does not use an image autoencoder, VQ codebook, Transformer, or joint perception/dynamics fine-tuning.

The original encoder uses ReLU and outputs directly in physical units. An exploratory repair uses LeakyReLU with slope 0.01 and $\hat{y} = \mu_{\text{train}} + \sigma_{\text{train}} r_\theta$, retaining width, depth, loss, and training budget. All six encoders are retrained under this rule. Changing activation and output parameterization together prevents isolating their individual causal contributions.

5.2. Road-relative dynamics

For an arc-length centerline within its valid coordinate chart, slip-free kinematics gives

$$\dot{s} = \frac{v \cos \psi}{1 - d\kappa}, \quad (2)$$

$$\dot{d} = v \sin \psi, \quad (3)$$

$$\dot{\psi} = \omega - \kappa \dot{s}. \quad (4)$$

The independently predicted curve and legacy map labels only approximate these arc-length identities; the original and guarded models are structured predictors rather than exact physical coordinate transformations. The consistent variants below explicitly address this distinction. The implementation adds learned lateral and heading residuals and per-step increments in speed and yaw rate. Three multilayer perceptrons, each with two 64-unit ReLU hidden layers, consume normalized $d, \psi, v, \omega, \kappa$ and actions. These actuation increments are not recovered physical bicycle parameters. The original variant uses Euler updates, interpolated curvature from the visual head, and a separate predicted centerline for reconstruction.

A cubic Hermite interpolant $P(s)$ uses centered interior tangents, an initial road- x tangent, and a final chord tangent. With tangent heading $\theta(s)$, the vehicle position is $P(s) + d[-\sin \theta(s), \cos \theta(s)]^\top$ and heading is $\theta(s) + \psi$. An ego transform removes initial translation and rotation. Outside the 4.5 m preview, the position readout extends its endpoint tangent while the independent curvature head holds its last value. These two extensions are not generally geometrically consistent.

Numerical safeguards. The guarded variant uses $\max(0.2, 1 - d\kappa)$, wraps heading, and bounds learned increments with hyperbolic tangents. Speed and yaw-rate bounds are training-only 99.5th percentiles of absolute one-frame changes, floored at 0.01 in their respective units. Lateral and heading residual limits are 0.1 m and 0.1 rad per step. Positive denominator protection extends the numerical update outside its valid chart; it does not make such states physically valid. We record states with $1 - d\kappa \leq 0$. Finite increments imply neither bounded states nor bounded prediction errors.

Geometric consistency. Nominal arc-length knots do not guarantee a unit-speed predicted interpolant. Let q be its actual parameter, $g = \|P'(q)\|$, and

$\kappa_P = \det(P'(q), P''(q))/g^3$. Consistent dynamics requires

$$\dot{q} = \frac{v \cos \psi}{g(1 - d\kappa_P)}, \quad \dot{\psi} = \omega - \kappa_P g \dot{q}. \quad (5)$$

We evaluate this correction with Euler and explicit midpoint integration, using the same curve in dynamics and readout. The independent curvature head is ignored, g is floored at 0.1, and the other safeguards are retained. Tangent extension uses $g = 1$ and $\kappa_P = 0$ outside the preview. Synthetic checks assess readout equivalence, finite-difference derivatives, and the coordinate-to-Cartesian velocity identity. Consistency does not imply that the predicted curve matches the real road.

5.3. Controls and training

A calibrated kinematic control learns seven global coefficients for throttle gain, drag, effective length, steering gain and bias, drive bias, and yaw-response lag. It receives initial speed, yaw rate, and future actions, without road inputs. Two Cartesian residual controls have the same architecture; one receives the initial road state and frozen visual preview, and the other zeros these inputs. Their residual network has two 96-unit ReLU hidden layers and five outputs. Its inputs are scaled planar position, sine/cosine of heading, normalized speed and yaw rate, two actions, and 32 normalized road-context quantities. Both learn residuals to analytic planar kinematics. These implementations are not claimed reproductions of DVBF, GOKU-net, or Vid2Param. Constant speed and yaw rate provide an additional action-independent diagnostic.

Perception training uses 25 epochs, Adam with initial learning rate 10^{-3} , cosine decay, batch size 64, and frame stride 2. The checkpoint minimizes validation normalized supervision loss. Dynamics use 40 epochs, 32-step windows, batch size 128, stride 4, and a common physical-output loss. Initial learning rates are 10^{-3} for neural dynamics and 10^{-2} for calibrated kinematics, with cosine decay. Gradient norms are clipped at 1 for dynamics and 5 for perception. Dynamics checkpoints minimize validation 100-step endpoint error. All seeds (0, 1, 2) are retained.

The loss averages squared errors in ego-frame x, y , wrapped heading, speed, and yaw rate, scaled by 0.5 m, 0.5 m, 0.5 rad, and training standard deviations of speed and yaw rate (floored at 0.01). Evaluation errors do not choose checkpoints within these runs; choices between modeling rounds are exploratory because earlier results had already been inspected.

6. Experimental Evaluation

6.1. Data and source-separated protocol

The physical archive contains two recordings with 5801 and 13597 frames at approximately 22 Hz, including images, steering/throttle, logged planar pose, speed, and a track map. Preprocessing yields 18 and 53 usable segments. These segments share acquisition conditions and layout; they are not independent sessions. Yaw rate uses at most five past yaw differences.

Direction A evaluates the first source, with 41 training, 11 validation, and one purged segment from the second source. Direction B evaluates the second source, with 13 training, 4 validation, and one purged segment from the first. Validation is the final 20% of ordered development-source segments; at least 115 original frames separate training and validation. Evaluation uses 844 and 1653 overlapping windows with stride 4 and horizon 100 (approximately 4.55 s).

Learned parameters and normalization statistics use training data only. Each round freezes its protocol, implementation hashes, and required input checkpoints before fitting; evaluation waits for all prescribed runs. Audits check data and checkpoint hashes, validation selection, window identity, recomputed metrics, and zero initial position error.

Both recordings were previously used in development. Retained track preprocessing also includes orientation chosen using the first recording and historically explored grid settings. Separation therefore concerns present fitting and checkpoint selection, not a prospectively untouched complete pipeline. No unseen-track or new-recording generalization claim follows.

Operating regimes and identifiability. Across retained frames, median speeds are 0.434 and 0.774 m/s in the two records. The first has constant throttle 0.566667, whereas the second ranges from 0.633333 to 0.766667. Source separation therefore also probes changes in operating regime. In a model $\dot{v} = k_a u - k_d v + b$, constant $u = c$ identifies the combination $k_a c + b$, not k_a and b separately: replacing them by $k_a + \Delta$ and $b - c\Delta$ leaves the update unchanged. We do not interpret fitted coefficients as uniquely recovered physical parameters. This limitation does not by itself explain all prediction errors.

6.2. Metrics

Endpoint error is

$$E_{xy}(k) = \frac{1}{N} \sum_{i=1}^N \|\hat{p}_{i,k} - p_{i,k}\|_2. \quad (6)$$

We report steps 25, 50, and 100, plus mean error over steps 1–100 (ADE), in meters. Mean $\pm$ standard deviation summarizes three training seeds, not uncertainty across independent recording sessions. Overlapping windows are not independent trials for significance testing. The primary metric is 100-step endpoint error.

6.3. Earlier experiments

Conference CartPole, Lunar Lander, and DonkeyCar-simulator results are retained with attribution [3]. Their supervision and architectures differ from the current road encoder. CarRacing was a development environment, but this article has no completed quantitative CarRacing validation of the road-context model. Earlier five-fold physical-car results are preserved in Appendix Appendix A, under their original information and selection conditions; they cannot be pooled with the new source-separated estimates.

7. Results

7.1. Initial controlled comparison

Table 1 shows lower mean Frenet endpoint error than the road-conditioned Cartesian residual control in both directions, but calibrated kinematics is better in both. These findings concern the specified implementations and budgets, rather than all possible models in either coordinate system.

Replacing both curvature and shape by a straight road raises Frenet endpoint error to 2.129 ± 0.546 and 2.415 ± 0.071 m. Setting curvature alone to zero while keeping shape gives 0.994 ± 0.126 and 1.517 ± 0.069 m. The joint intervention therefore does not isolate curvature-dependent dynamics. Inference-time interventions also introduce distribution shift; separately trained ablations provide a complementary comparison.

Table 1: Initial retrospective comparison. Errors in m; mean $\pm$ sample SD over three seeds. A and B evaluate the first and second original recordings.

Model	A: E_{100}	A: ADE	B: E_{100}	B: ADE
Calibrated kinematics	0.936 ± 0.005	0.455 ± 0.002	1.158 ± 0.006	0.595 ± 0.002
Cartesian, no road	1.230 ± 0.128	0.654 ± 0.082	1.466 ± 0.156	0.720 ± 0.049
Cartesian, road	1.344 ± 0.223	0.679 ± 0.063	1.653 ± 0.035	0.692 ± 0.014
Frenet	1.086 ± 0.150	0.608 ± 0.112	1.463 ± 0.037	0.688 ± 0.022

Table 2: Exploratory numerical safeguards and retrained ablations using the original frozen encoders.

Model	A: E_{100}	A: ADE	B: E_{100}	B: ADE
Guarded, full	1.022 ± 0.154	0.473 ± 0.062	1.432 ± 0.019	0.672 ± 0.025
Guarded, shape only	0.846 ± 0.048	0.400 ± 0.046	1.471 ± 0.062	0.688 ± 0.025
Guarded, no road	1.029 ± 0.078	0.492 ± 0.012	1.579 ± 0.161	0.675 ± 0.058
Guarded, oracle	0.933 ± 0.068	0.430 ± 0.035	1.285 ± 0.039	0.618 ± 0.020

7.2. Numerical stability and retrained ablations

Two original Frenet runs deteriorate after their validation-optimal epochs: final validation errors reach 99.875 and 21.681 m, with maximum pre-clipping gradient norms above 10^{12} and 10^{13} . Selected checkpoints reproduce the initial comparison exactly in the diagnostic rerun. Guarded updates avoid these extreme failures in the same runs, but can still leave the valid Frenet chart.

Table 2 reports models refit with fixed perception. Shape-only sets curvature to zero throughout fitting and prediction. No-road additionally replaces shape by a straight line and zeros initial road-relative displacement and heading; it does not isolate shape alone. The oracle receives true curvature and shape during fitting and evaluation; it is not a theoretical upper bound.

Shape-only improves on the full model in A but worsens it in B. Neither the safeguards nor the oracle establishes a consistent advantage over calibrated kinematics, so the remaining gap cannot be attributed solely to perception.

7.3. Geometric consistency and integration

A validation-only audit finds that predicted curves are not generally unit-speed in their nominal parameter. For one encoder, 80.1% of sampled locations have $|g - 1| > 0.2$. Independent and shape-derived curvatures

Table 3: Exploratory geometry and integration controls with the original encoders.

Model	A: E_{100}	A: ADE	B: E_{100}	B: ADE
Consistent, Euler	1.141 ± 0.051	0.537 ± 0.035	1.314 ± 0.034	0.611 ± 0.029
Consistent, midpoint	1.267 ± 0.361	0.571 ± 0.138	1.325 ± 0.042	0.609 ± 0.035
Body, midpoint	1.009 ± 0.056	0.486 ± 0.078	1.793 ± 0.221	0.775 ± 0.069

Table 4: Matched dynamics with six uniformly retrained conditioned encoders. All dynamics are refit from scratch; all variants and seeds are retained.

Model	A: E_{100}	A: ADE	B: E_{100}	B: ADE
Guarded, full	0.951 ± 0.089	0.435 ± 0.035	1.387 ± 0.038	0.672 ± 0.010
Guarded, shape only	0.870 ± 0.108	0.375 ± 0.058	1.394 ± 0.026	0.663 ± 0.022
Consistent, midpoint	1.020 ± 0.088	0.469 ± 0.046	1.300 ± 0.068	0.616 ± 0.044
Cartesian, road	1.402 ± 0.113	0.667 ± 0.041	1.584 ± 0.040	0.680 ± 0.029

also disagree. Even the coarse oracle interpolant is not exactly unit-speed. Synthetic checks verify the coordinate velocity identity in Eq. (5). On a synthetic circle, zero-residual 100-step endpoint error falls from 0.001293 m with Euler to 0.000119 m with midpoint integration. This checks the numerical implementation, not accuracy on recorded driving.

Table 3 evaluates consistent dynamics with the original encoders. The body midpoint control uses the same network structure and integrator, but clears road inputs. Geometry helps relative to this control in B; A has the opposite ordering. Calibrated kinematics still gives lower mean endpoint error in both directions.

7.4. Visual conditioning and matched dynamics

Training and validation logs identify one encoder with nearly constant outputs. Its best normalized validation loss is 0.8384, versus 0.1220 and 0.1280 for the other seeds in that direction. This diagnoses representation collapse, not its precise optimization mechanism. Uniform conditioning repair improves validation loss in five of six runs; one changes from 0.1220 to 0.1371. The collapsed run improves to 0.2953 with nonconstant outputs. Validation improvements alone do not demonstrate improved trajectory generalization. Table 4 uses the same previously examined sources. The consistent model ignores the independent curvature head, while shape-only clears that head during dynamics. Neither ablation isolates curvature supervision of the jointly

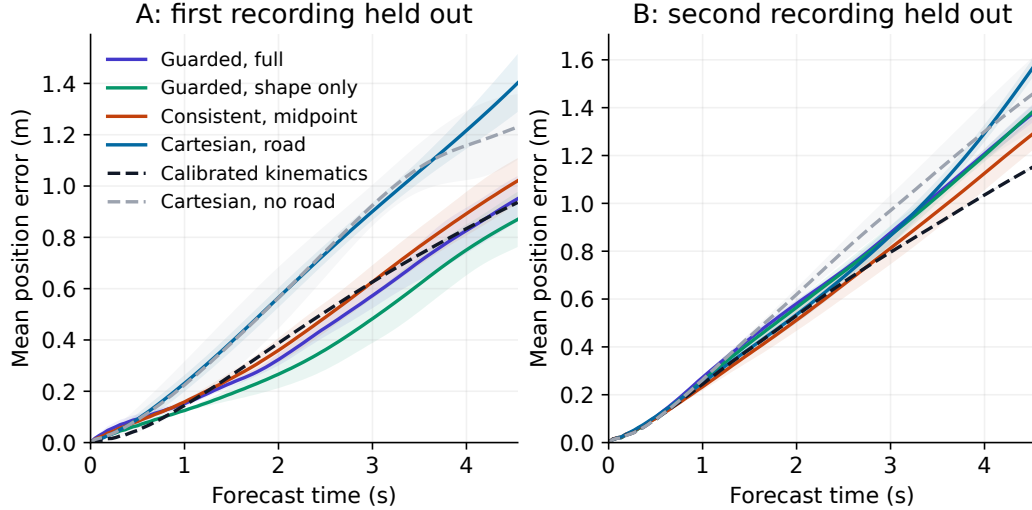


Figure 2: Complete conditioned-dynamics comparison and fixed controls. Lines are means over three training seeds; shading is sample standard deviation across seeds, not a confidence interval over recording sessions. All curves use the same source-specific windows, privileged initialization, and recorded actions. Both sources had been inspected in earlier development.

trained encoder. The kinematic and no-road Cartesian references in Table 1 are unaffected by encoder changes.

7.5. Purged temporal prediction within both sources

To distinguish source/regime transfer from forecasting within recorded operating conditions, we add an exploratory chronological split separately within each source. The first approximately 60% of ordered segments train the models, the next 20% select checkpoints, and the last 20% evaluate them. Boundary segments are purged to leave at least 115 original frames between adjacent partitions. This produces 9+30 training segments, 3+10 validation segments, 4+11 evaluation segments, and 2+2 purged segments. All encoders and six dynamics variants are trained from scratch using the same 25/40-epoch budgets. The guarded full model and pooled-window endpoint metric are fixed before fitting; no family is selected retrospectively as the winner.

This experiment was designed after inspecting the earlier studies and the recorded operating ranges. It does not provide fresh independent confirmation, and it does not replace the source-held-out stress tests. The narrower question

Table 5: Exploratory chronological holdout with both sources represented during training; later operating shifts may remain. Pooled-window E100 is primary; the last two columns expose the original sources separately. Errors in m, mean and sample SD over three seeds.

Model	Pooled E_{100}	Pooled ADE	Rec. 1 E_{100}	Rec. 2 E_{100}
Calibrated kinematics	0.898 ± 0.006	0.437 ± 0.004	0.858 ± 0.006	0.913 ± 0.005
Cartesian, no road	0.930 ± 0.335	0.367 ± 0.081	0.675 ± 0.355	1.029 ± 0.329
Cartesian, road	1.095 ± 0.212	0.424 ± 0.090	0.793 ± 0.310	1.211 ± 0.175
Guarded, full	0.343 ± 0.010	0.197 ± 0.005	0.208 ± 0.028	0.395 ± 0.004
Guarded, shape only	0.412 ± 0.018	0.246 ± 0.015	0.260 ± 0.019	0.471 ± 0.026
Consistent, midpoint	0.372 ± 0.028	0.208 ± 0.012	0.301 ± 0.042	0.399 ± 0.023

is prediction on later portions of the same sources, with both sources represented during training. Training throttle nevertheless contains only 0.566667 and 0.633333, whereas the second full record reaches 0.766667. This split is not asserted to be identically distributed or to cover every later operating condition. Table 5 reports the pooled primary metric and both source-specific endpoint errors. The prespecified guarded full model gives a pooled endpoint error of 0.343 m, compared with 0.898 m for calibrated kinematics and 1.095 m for the road-conditioned Cartesian control. These values must be interpreted together with the per-source results and the cross-source stress tests, not as evidence for unseen-track generalization.

7.6. A single unit-speed road representation

The preceding consistency audit motivates an exploratory encoder whose curvature and position describe one curve by construction. Ten predicted curvature knots replace the independent curvature and shape heads. For the nine intervals of length $\ell = 0.5$ m, let $\bar{\kappa}_j = (\hat{\kappa}_j + \hat{\kappa}_{j+1})/2$ and $\theta_j = \sum_{i < j} \bar{\kappa}_i \ell$. At distance $u \in [0, \ell]$ in interval j ,

$$C(q_j + u) = C(q_j) + u \operatorname{sinc}(\bar{\kappa}_j u/2) \begin{bmatrix} \cos(\theta_j + \bar{\kappa}_j u/2) \\ \sin(\theta_j + \bar{\kappa}_j u/2) \end{bmatrix}, \quad (7)$$

$$\theta(q_j + u) = \theta_j + \bar{\kappa}_j u, \quad (8)$$

where $\operatorname{sinc}(x) = \sin(x)/x$, extended continuously at zero. Position and tangent are continuous; curvature can jump at interval boundaries. The curve has unit parameter speed, and the same interval curvature drives midpoint Frenet

Table 6: Exploratory coupled representation on the unchanged chronological split. Mean and sample SD over all three seeds; all variants retained.

Model	Pooled E_{100}	Pooled ADE	Rec. 1 E_{100}	Rec. 2 E_{100}
Coupled arc, full	0.400 ± 0.005	0.217 ± 0.001	0.260 ± 0.008	0.454 ± 0.005
Coupled arc, shape only	0.454 ± 0.042	0.259 ± 0.026	0.260 ± 0.040	0.529 ± 0.053
Arc, independent encoder	0.468 ± 0.022	0.235 ± 0.009	0.314 ± 0.021	0.528 ± 0.023
Cartesian, coupled road	1.080 ± 0.021	0.414 ± 0.022	0.722 ± 0.051	1.218 ± 0.035

updates and position readout. Outside the preview, tangent extension has zero curvature. This is a consistent implementation of standard curve geometry, not a new Frenet identity or a stability guarantee.

The encoder retains the conditioned CNN backbone and is jointly supervised by legacy curvature and shape targets; shape gradients propagate through the arc integral. All three seeds are refit for 25 epochs. Four dynamics variants are refit for 40 epochs on exactly the temporal split in Section 7.5. The full coupled model and pooled E_{100} are fixed before fitting. The independent-encoder control uses the preceding frozen CNN’s curvature output with the same arc decoder and dynamics. The shape-only variant removes curvature from the dynamics but retains the curved readout; it does not remove all curvature information. Cartesian dynamics receives the same coupled preview. The full coupled model has pooled endpoint error 0.400 m, compared with 0.468 m for the independent-encoder arc control and 1.080 m for Cartesian dynamics using the coupled preview. The unchanged calibrated-kinematic reference is 0.898 m. These comparisons jointly expose coupling, geometry, and model-family effects.

7.7. Compatibility of the geometric supervision

The persisted centerline is smoothed after resampling and is not reparameterized by arc length afterward. Its nominal grid length is 10.450 m, whereas the smoothed polygon length is 9.743 m. At the native 0.05 m grid, the median absolute deviation of parameter speed from one is 0.0512. Stored curvature is additionally smoothed. Thus these direct map-derived labels are not exact, mutually consistent geometric ground truth.

On the temporal validation windows, integrating the stored curvature with the piecewise-arc decoder differs from the stored shape by 0.207 m coordinate RMSE. This difference combines map parameterization, separate curvature smoothing, and coarse interval approximation; it does not isolate one cause.

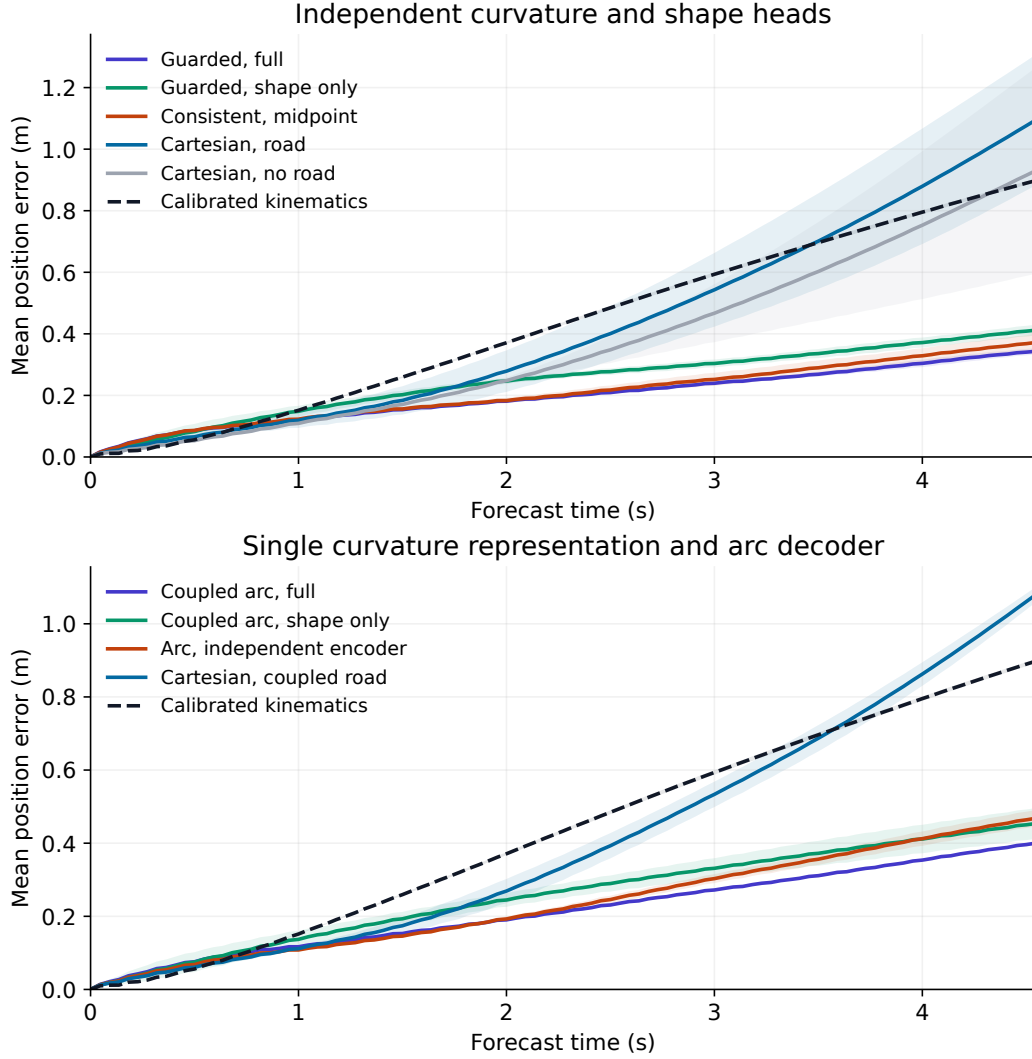


Figure 3: Chronological-holdout errors for every prescribed variant. The panels separate the independent-head and coupled-road suites; calibrated kinematics is repeated as a common reference. Means and shading summarize three seeds (sample standard deviation), not independent recording uncertainty.

Joint supervision of an exact unit-speed representation therefore introduces target mismatch. No test-window information is used in this audit, and the existing targets and complete experimental results are retained. This limits conclusions about the best attainable accuracy of geometrically consistent

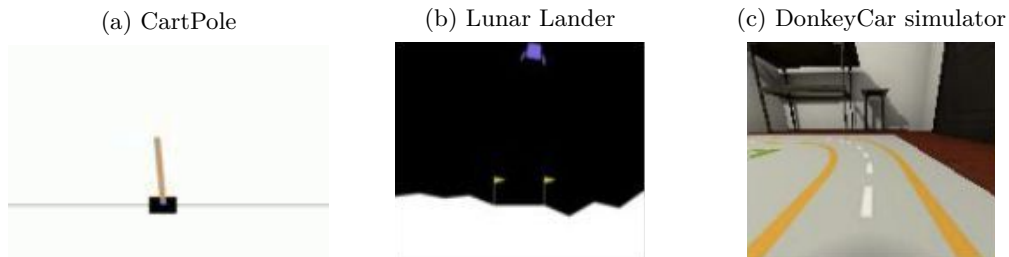


Figure 4: Original conference environments: CartPole, Lunar Lander and the DonkeyCar simulator. Frames are extracted from the ground-truth rows of the conference qualitative figures [3]; the simulated driving scene is distinct from the physical platform evaluated in the present study.

learning from corrected labels. For the fixed guarded full model, seed-averaged endpoint error improves on calibrated kinematics and both Cartesian controls in all 14 eligible temporal test segments. Only 14 of the 15 assigned segments are long enough for the prescribed 15-frame context and 100-step horizon, yielding 527 windows. Full dynamics improves on shape-only in 10 of those segments. This is descriptive paired evidence: overlapping windows and segments from two recordings do not provide 14 independent acquisition sessions. The maximum fraction of monitored full-step states with an invalid Frenet denominator in the guarded full model is 0.0569%. Finite rollouts therefore do not establish chart validity. Midpoint internal stages are not included in this monitoring.

7.8. Retained Conference Benchmark Results

We restore the original conference results [3] in Figures 5–9. These are inherited experiments, not additional runs of PIWM-Frenet. They compare intrinsic and extrinsic encoders with continuous or discrete latents over 30 prediction steps and supervision-noise levels $\delta \in \{0, 5\%, 10\%\}$. The conference protocol uses five-fold cross-validation. State RMSE in these benchmarks is distinct from the meter-valued position error of the physical-car evaluation.

Prediction and architectural choice. Figures 5, 6 and 7 retain all six panels for each environment. Extrinsic PIWM with discrete latents gives the lowest long-horizon error among the extrinsic variants shown. The intrinsic setting has a different ranking: continuous PIWM generally outperforms the discrete

variant. Thus the conference results support choosing the latent parameterization together with the encoder architecture, rather than a universal advantage for quantization. The driving encoder comparison in Table A.8 revisits this choice for the new road-perception task.

Parameter estimates and visual rollouts. Figure 8 restores the parameter comparison for CartPole and Lunar Lander. The reference lines make both agreement and departures from the known parameters visible as supervision noise changes. We omit the original DonkeyCar wheelbase panel from this recovery comparison because it has no known-parameter reference. Figure 9 retains all five comparison rows from the original visual rollout example, with three display columns selected for legibility. These examples illustrate visual behavior and do not replace the quantitative comparisons above. The original controller diagnostic is also retained in Appendix Appendix C.

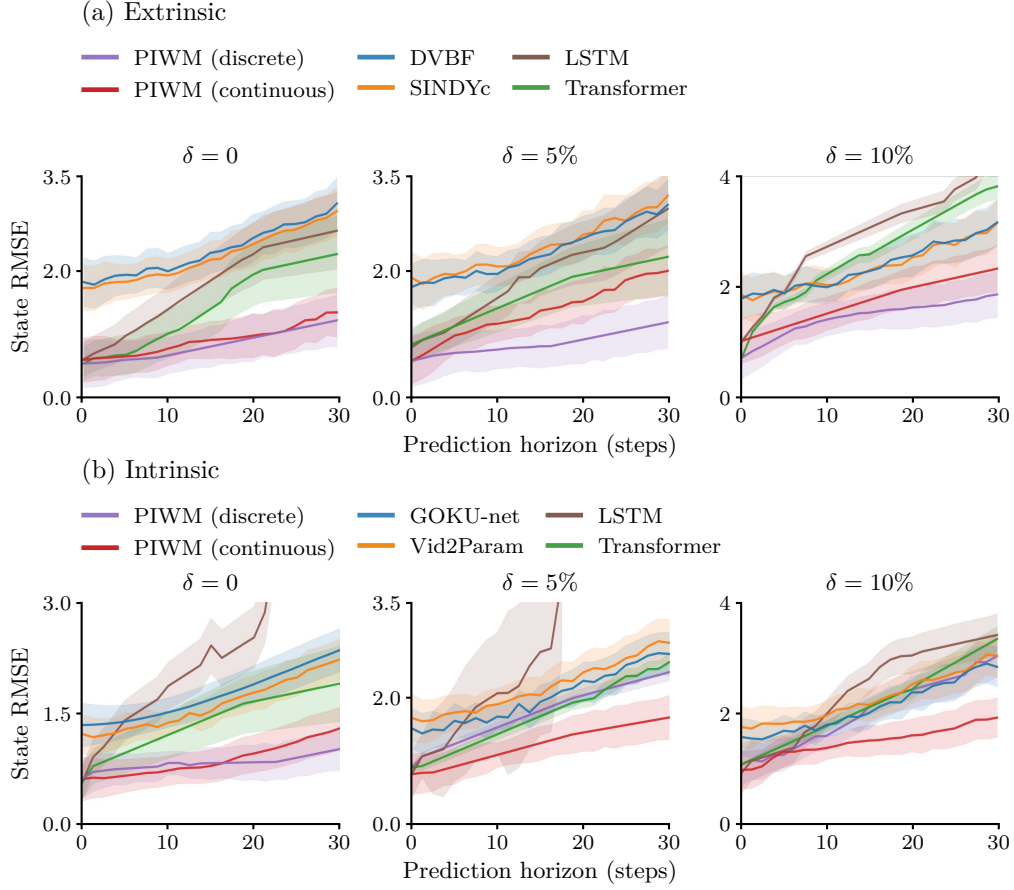


Figure 5: Conference CartPole results [3]. State RMSE over 30 steps for (a) extrinsic and (b) intrinsic methods, under three supervision-noise levels. Curves and shaded regions are the original embedded plot images; labels and layout have been reset. Panel-specific vertical limits are retained. The archived figure does not specify the statistical meaning of its shading.

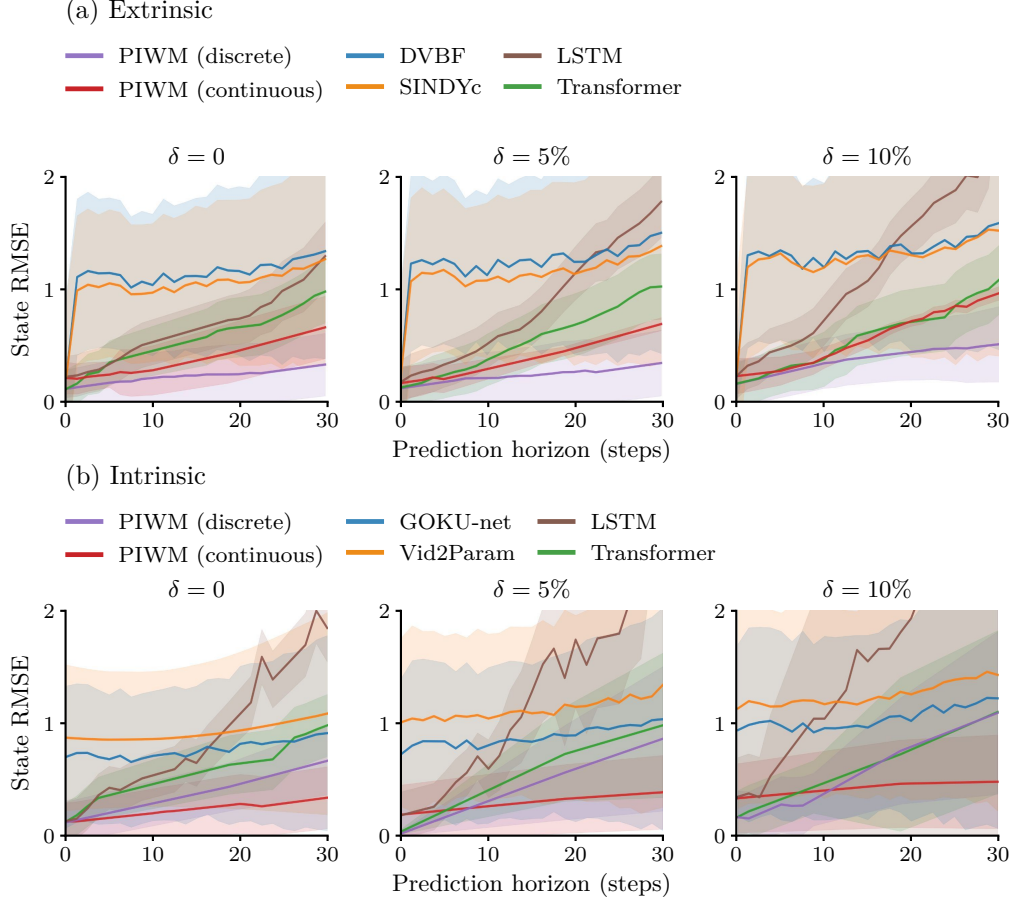


Figure 6: Conference Lunar Lander results [3]. The layout and method groups follow Figure 5. Original curves and shaded regions are preserved; all panels retain the source RMSE range $[0, 2]$.

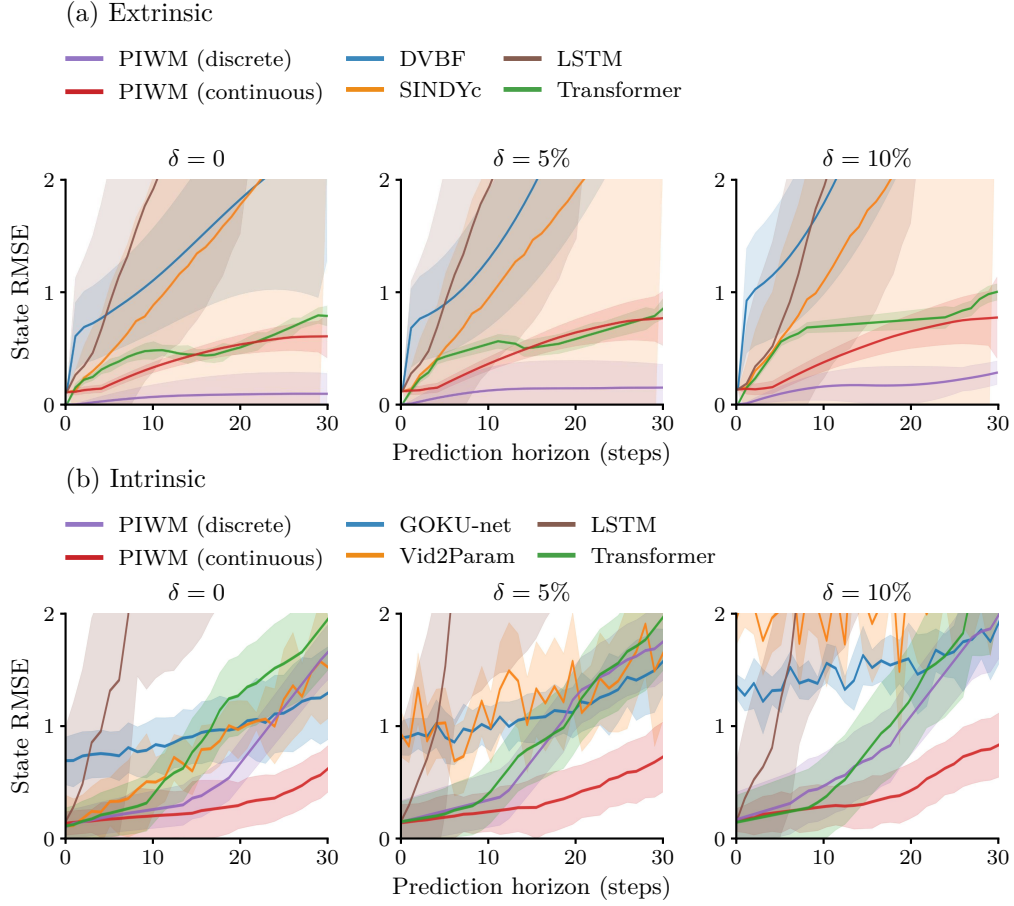


Figure 7: Conference DonkeyCar-simulator results [3]. The layout follows Figure 5. Original curves and shaded regions are preserved, including the source vertical limit of 2; curves beyond that limit are clipped as in the original. These are 30-step state-RMSE results and must not be read as the 100-step position errors of the physical car.

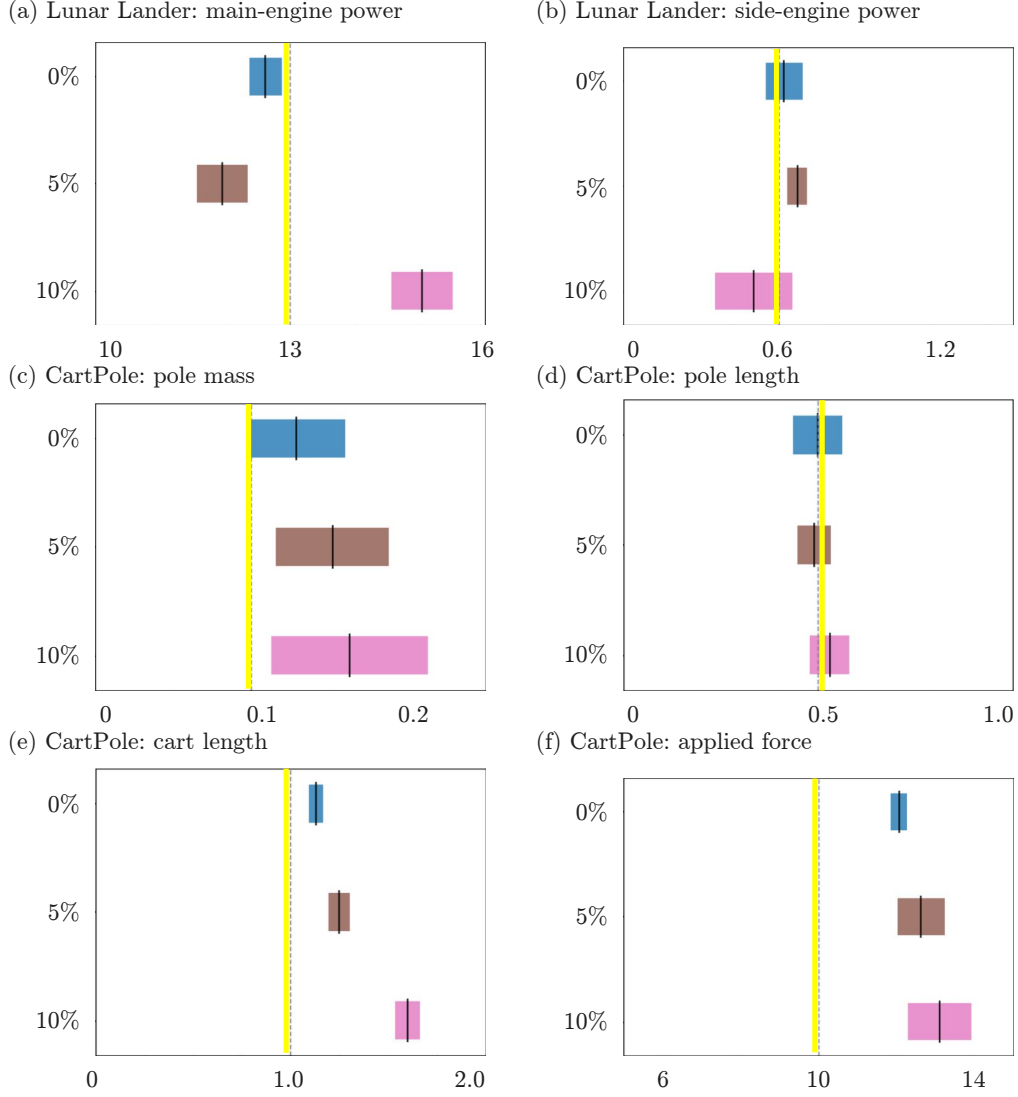


Figure 8: Conference parameter estimates for the two controlled benchmarks [3]. Each panel retains the original colored intervals, center marks and yellow ground-truth reference. Rows correspond to $\delta = 0, 5\%, 10\%$, from top to bottom. Parameter names and scales follow the source figure; the source does not specify the interval statistic.

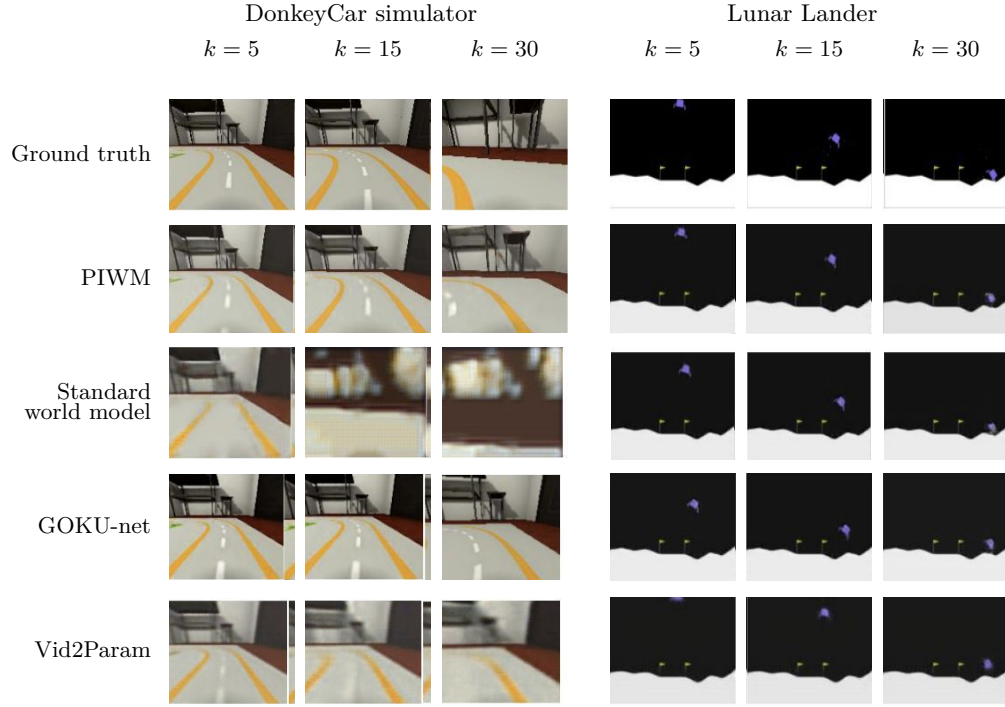


Figure 9: Conference visual rollouts at $\delta = 5\%$ [3]. Ground truth and the four original model rows are shown at $k = 5, 15, 30$ for the DonkeyCar simulator and Lunar Lander. Frames are extracted without retouching; the intermediate display columns and the slide footer have been omitted.

8. Discussion

The studies distinguish physical meaning, numerical behavior, and predictive accuracy. The first two do not imply the third. Named variables expose coordinate failures and inconsistent geometric representations, as the audits demonstrate. Accuracy benefits depend on the recording, split, baseline, and implementation. The chronological-holdout gains are substantial and reproducible from saved weights, but the protocol was designed during development on reused sources. They support within-source prediction under the stated information conditions, while the source-separated results expose limited transfer.

Road geometry alone does not alter action-conditioned planar kinematics. As context for learned actuation, it may encode correlations with driving conditions or the behavior generating the data. These experiments do not identify such correlations as causal effects. Re-indexing a fixed preview removes one learned recurrent update but still propagates errors in progress and geometry.

The principal limitation is two reused recordings on one track. Seeds assess optimization variability, not variation across independent tracks, vehicles, or acquisition sessions. Successive repairs were informed by previous analysis; their evaluation is exploratory even with validation-only checkpoint selection. Fresh acquisition is needed for prospective confirmation of a selected method. Other limitations are privileged initialization, map-derived encoder labels, recorded future actions, finite preview, and absence of closed-loop assessment. Numerical guards can remain finite outside the valid coordinate chart. No safety or general bounded-error guarantee follows.

The direct road encoder differs from the conference autoencoding pipeline, whose weak-supervision results do not establish weakly supervised road perception. Historical reduced baselines also cannot establish superiority over the full published methods. A deployment accuracy claim requires benefits over calibrated dynamics across independent conditions and evaluation with an initial-state estimator. Favorable subsets of these development experiments cannot satisfy those requirements.

9. Conclusion

We developed and audited a camera-derived road-context extension of PIWM for offline action-conditioned prediction. Explicit geometry makes

coordinate assumptions and failures inspectable. Controlled studies distinguish input information, reconstruction, numerical safeguards, and dynamics training. The original Frenet predictor does not beat calibrated kinematics in the two source-held-out directions. The guarded model substantially improves chronological prediction over matched controls, with improvements in both recordings and across all eligible test segments against the three principal baselines. This contrast with source-transfer performance identifies a useful but limited operating regime for the representation; geometric consistency alone is insufficient to establish predictive superiority. The evidence supports a limited empirical account of the representation, not resolved partial observability, camera-only deployment, or guaranteed stability.

Declaration of Competing Interest

[AUTHOR CONFIRMATION REQUIRED: competing-interest declaration.]

CRedit Authorship Contribution Statement

[CRedit ROLES — TO BE COMPLETED BEFORE SUBMISSION.]

Data Availability

[AUTHOR CONFIRMATION REQUIRED: data/code release location, permissions, and access restrictions.]

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Declaration of Generative AI Use

[AUTHOR REVIEW REQUIRED BEFORE SUBMISSION.] GPT-6 (Codex) assisted with code inspection, experimental implementation, analysis, and manuscript revision. The final disclosure requires author verification of its scope and compliance with the journal policy. This draft does not assert that the authors have already verified or approved all assisted outputs.

Appendix A. Historical Physical-Car Development Results

These tables and figures preserve earlier five-fold development results. The 71 segments come from the same two recordings. Displayed evaluation windows also selected dynamics checkpoints, so these estimates are optimistic as measures of generalization. The original Vid2Param visual encoder was shared across folds; DVBF and GOKU-net were reduced to dynamics-only variants. Budgets and information pathways differed. These are descriptive results under the original protocol, not a clean ranking of published methods.

The 0.463 m result uses surveyed-map position reconstruction and privileged initialization. The 0.516 m row uses perceived-shape readout with the original selection and initialization conditions. Differences from the source-separated study also involve splits, losses, horizons, and readout implementation; they are not a controlled estimate of one isolated source of optimism. These results motivated follow-up studies. Small effects from curvature replacement and gains under some noise conditions do not identify a unique causal mechanism; they should be read alongside the controlled main-text results.

Table A.7: Historical five-fold development errors (m). Displayed windows also selected checkpoints. Frenet uses map-assisted readout except in the map-free row; all rows use privileged initialization. Baseline names refer to local reduced implementations. These are not independent test estimates.

Model	$E_{xy}(25)$	$E_{xy}(50)$	$E_{xy}(100)$
Vid2Param [56]	0.036 ± 0.008	0.107 ± 0.008	0.703 ± 0.142
GOKU-net [38]	0.046 ± 0.012	0.131 ± 0.013	0.725 ± 0.107
DVBF [40]	0.061 ± 0.011	0.173 ± 0.031	0.837 ± 0.131
SINDYc [57]	diverges in all 2483 windows of all five folds		
PIWM-Frenet (ours, perceived κ)	0.141 ± 0.009	0.245 ± 0.021	0.463 ± 0.045
<i>PIWM-Frenet (map-free)</i>	0.146 ± 0.012	0.258 ± 0.023	0.516 ± 0.056

Table A.8: Historical curvature-source substitutions using the original dynamics and evaluation protocol. Curvature RMSE is in inverse meters; endpoint error is in meters. Camera refers to the curvature input only; readout and initialization retain the original privileged information. Limited sensitivity here does not establish absence of error propagation.

Source of κ	Input	RMSE $_{\kappa}$	$E_{xy}(100)$
Known track map	map + pose	—	0.505 ± 0.043
Ground-truth profile	true κ	—	0.466 ± 0.046
Extrinsic autoencoder	camera	0.193	0.454 ± 0.048
LSTM	camera	0.898	0.456 ± 0.089
Intrinsic autoencoder	camera	0.136	0.460 ± 0.045
CNN, frozen backbone	camera	0.247	0.461 ± 0.053
CNN, warm-started backbone	camera	0.148	0.462 ± 0.052
CNN, from scratch	camera	0.154	0.463 ± 0.045
VQ + Transformer	camera	0.219	0.463 ± 0.051
Transformer	camera	0.122	0.471 ± 0.047

Table A.9: Historical perturbation of dynamics supervision, mean and standard deviation over five development folds. Road-encoder labels are not perturbed, so this does not establish weakly supervised visual road learning. The Gaussian row is a matched-magnitude diagnostic, not evidence about all noise distributions.

Model	$\delta = 0\%$	$\delta = 5\%$	$\delta = 10\%$
PIWM-Frenet (ours)	0.463 ± 0.045	0.452 ± 0.035	0.319 ± 0.044
<i>PIWM-Frenet (map-free)</i>	0.516 ± 0.056	0.506 ± 0.049	0.390 ± 0.036
Vid2Param	0.703 ± 0.142	0.919 ± 0.189	0.938 ± 0.211
GOKU-net	0.725 ± 0.107	0.896 ± 0.152	0.864 ± 0.142
DVBF	0.837 ± 0.131	1.067 ± 0.188	1.075 ± 0.177
<i>matched-magnitude Gaussian control, ours at $\delta = 10\%$: 0.313 ± 0.041</i>			

Table A.10: Historical initial-state perturbations with ten draws per window. Values are preserved from the original aggregation. The clean column differs slightly from the main historical table and should not be treated as an exact reproduction. Noise is relative to the original state-normalization scales.

Model	$\sigma = 0$	0.25	0.5	1.0	1.5
PIWM-Frenet (ours)	0.462	0.485	0.540	0.685	0.823
Vid2Param	0.699	0.726	0.788	0.977	1.199
GOKU-net	0.724	0.768	0.856	1.097	1.503
DVBF	0.840	0.876	0.974	1.267	1.681

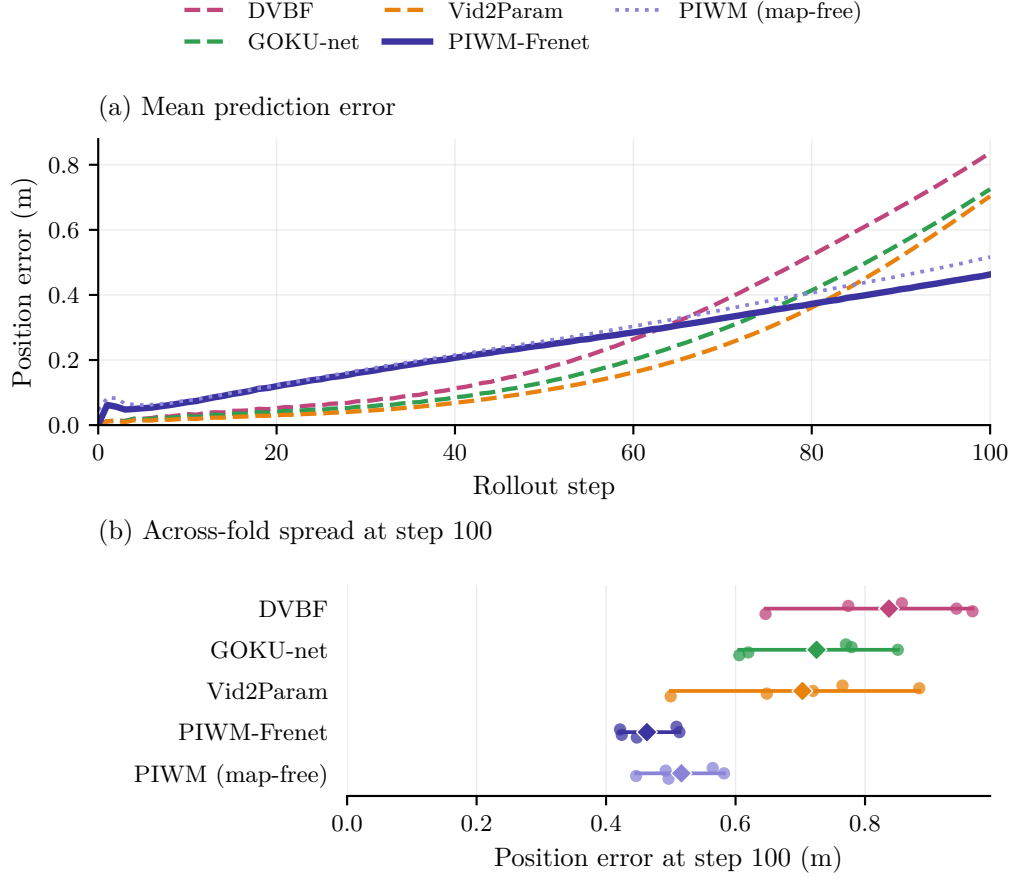


Figure A.10: Historical five-fold development curves with ground-truth initialization and checkpoint selection on the displayed windows. The principal Frenet curve uses map-assisted reconstruction. SINDYc is omitted after divergence.

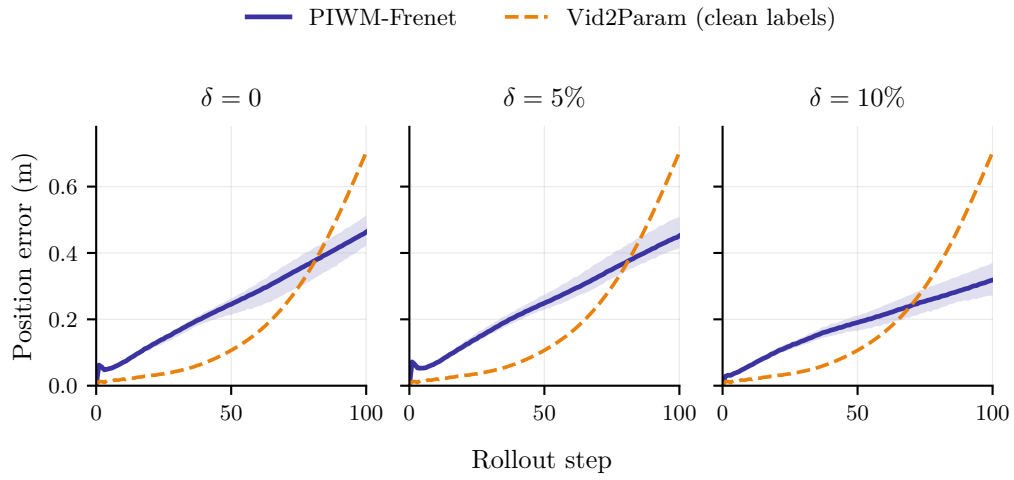


Figure A.11: Historical dynamics-supervision perturbation curves. Solid curves average five development folds; shading spans their range. The repeated dashed reference uses clean-label Vid2Param. This is not noisy road-encoder supervision.

Appendix B. Weak Supervision Noise Model

The weak supervision in retained conference benchmark experiments follows the noise model from the conference paper. For the i -th state component with valid range size $\mathcal{X}_i$, we generate a randomly shifted center $\tilde{\mu}_i = x_i + \Delta_i$, $\Delta_i \sim \text{Unif}[-\frac{1}{2}\delta|\mathcal{X}_i|, \frac{1}{2}\delta|\mathcal{X}_i|]$, and draw the supervisory distribution as $p_i(x) = \text{Unif}[\tilde{\mu}_i - \frac{1}{2}\delta|\mathcal{X}_i|, \tilde{\mu}_i + \frac{1}{2}\delta|\mathcal{X}_i|]$. At each timestep, $L = 50$ samples are drawn to form the proxy supervision set Ξ_t .

The physical-car historical study in Table A.9 perturbs dynamics state supervision and initialization. It does not perturb the image-to-curvature or image-to-shape training labels. The new controlled studies use direct map-derived labels; their road preview is inferred from images, while initial road-relative state remains privileged.

Appendix C. Conference Controller Diagnostic

The conference study [3] passed reconstructed observations through the original fixed controller to assess task-relevant visual information. Table C.11 reproduces the archived results, separately from trajectory-prediction error. The continuous intrinsic variant gives the best controller agreement in these examples; the strongest prediction variant need not have the strongest reconstruction-based control metric.

Table C.11: Conference controller results on reconstructed observations, reproduced from [3]. Entries retain the source’s reported $\pm$ quantities; the archived table does not define their dispersion statistic. Accuracy entries are percentages. These results do not evaluate the new Frenet model.

Architecture	Latent	$\delta = 0$	$\delta = 5\%$	$\delta = 10\%$
<i>DonkeyCar simulator: action RMSE $\downarrow$</i>				
Intrinsic	Continuous	0.12 ± 0.04	0.13 ± 0.04	0.15 ± 0.05
Intrinsic	Discrete	0.21 ± 0.15	0.29 ± 0.16	0.32 ± 0.20
Extrinsic	Continuous	0.15 ± 0.05	0.16 ± 0.05	0.22 ± 0.06
Extrinsic	Discrete	0.15 ± 0.04	0.17 ± 0.05	0.19 ± 0.05
<i>Lunar Lander: action accuracy (%) $\uparrow$</i>				
Intrinsic	Continuous	93.0 ± 1.8	90.5 ± 2.0	87.1 ± 2.2
Intrinsic	Discrete	85.5 ± 2.5	82.1 ± 2.8	78.3 ± 3.1
Extrinsic	Continuous	86.2 ± 2.4	83.5 ± 2.6	80.0 ± 2.9
Extrinsic	Discrete	91.5 ± 2.1	88.6 ± 2.3	84.5 ± 2.5
<i>CartPole: action accuracy (%) $\uparrow$</i>				
Intrinsic	Continuous	98.0 ± 1.0	96.5 ± 1.2	94.0 ± 1.5
Intrinsic	Discrete	95.0 ± 1.6	91.5 ± 2.0	87.2 ± 2.5
Extrinsic	Continuous	95.5 ± 1.5	92.0 ± 1.8	88.0 ± 2.2
Extrinsic	Discrete	97.2 ± 1.1	95.0 ± 1.4	92.5 ± 1.8

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